Rubén Salas Alcornada

ruben.sls@outlook.com · +34 617 69 19 08 · rubensalas.ai · github.com/r-salas · Madrid

ML Engineer specializing in production LLM-powered conversational and voice AI. 9 years building multichannel assistants (voice, WhatsApp, web) at scale, from Aura, Telefónica's AI assistant, to leading the technical architecture of a SaaS platform built from the ground up. The platform now serves 50+ paying customers and resolves 70% of conversations without human intervention.

EXPERIENCE

Technical Lead · Canaia · 2024 - Present

Early-stage conversational AI startup (omnichannel SaaS for SMBs and custom projects for enterprise clients). Joined as one of the first two engineers and took ownership of the product's technical architecture.

- Designed and built the LLM-powered assistant engine from the ground up, now in production with 50+ paying customers across phone, WhatsApp Business API, web and social media, resolving 70% of conversations without human handoff.
- Built real-time voice pipelines (STT, LLM, TTS) on Deepgram, ElevenLabs and Twilio, reducing cost and latency through caching, dynamic model routing and multi-provider failover.
- Implemented function calling so assistants can take real-world actions: booking management, calendar sync and third-party system integrations.
- Led the development of Text-to-SQL assistants over 20+ table schemas for a global industrial vehicle manufacturer, a leading media group and an agency serving IBEX 35 companies, with RBAC and query validation guardrails.
- Built an LLM-based conversation analytics system for automated quality control in production, and delivered native speech-to-speech PoCs and agent-assist copilots for support teams.
- Define the architecture and tech stack for a 5-person engineering team, run technical interviews and act as the technical point of contact for clients.

Machine Learning Engineer · Teleperformance - MAIA (AI department) · 2022 - 2024

Teleperformance Spain's AI department (8-person team), building customer service virtual assistants for enterprise and multinational clients.

- Acted as the team's technical lead, defining the architecture and tech stack for the department's assistants, including the microservices design and communication flows between components.
- Took multichannel virtual assistants to production for enterprise clients, including the department's largest: a Rasa-based voice assistant for a telecom operator with 1M+ customers.
- Built an end-to-end call analytics pipeline for a multinational home appliance manufacturer: automated call transcription, LLM-generated summaries, service metrics and direct integration with the client's CRM.
- Drove early LLM adoption in 2023, building the department's first prototypes with LLMs and function calling ahead of the shift away from classic NLU architectures (Rasa).

Machine Learning Engineer · Telefónica - Programmatic Advertising · 2021 - 2022

- Trained audience segmentation and interest prediction models that decide in real time which ad creative each user sees, using distributed clustering and classification algorithms.
- Processed and analyzed 30M+ record datasets with PySpark, and built campaign performance dashboards used by business teams.

Machine Learning Engineer · Telefónica - Digital Home Prototyping (Aura) · 2017 - 2021

7-person team developing Aura, Telefónica's AI assistant.

- Prototyped new conversational features for Aura during its early stage, including its integration into TV (Movistar Home).
- Deployed PyTorch NLP models (transformers/BERT and embeddings) to production for text classification and intent recognition, and built and deployed REST APIs on AWS.
- Built the computer vision and audio classification prototype (PyTorch, YOLO) for detecting people, pets and sounds in home security cameras. The resulting feature launched commercially in Movistar Prosegur Alarmas and reached 10,000+ customers in its first three months.

PROJECTS

AI Voice Assistant for my Resume · rubensalas.ai · github.com/r-salas/website

In-browser voice agent that answers questions about my professional background. Real-time streaming STT, LLM, TTS pipeline over WebRTC, self-hosted and deployed with infrastructure as code (Terraform). Try it and ask it about my experience.

Complex Document Parsing to Markdown with Vision LLMs · github.com/r-salas/markscribe

Converts documents with complex layouts (tables, charts, multi-column pages) into structured Markdown using vision LLMs, generating text descriptions of visual content such as charts and figures. Designed as the ingestion layer for RAG systems.

SKILLS

Voice & Conversational AI

Real-time STT, LLM, TTS pipelines · Deepgram · ElevenLabs · Cartesia · Azure Speech · OpenAI Realtime API · Twilio · WebRTC/WebSockets · WhatsApp Business API · Rasa · NLU & intent recognition

LLMs & AI Agents

OpenAI · Anthropic · Gemini · OpenRouter · RAG · Qdrant · pgvector · Function calling · MCP · OpenAI Agents SDK · LangChain/LangGraph · Text-to-SQL · Guardrails

ML & Data

Python · PyTorch · Transformers/BERT · scikit-learn · Computer Vision (YOLO) · PySpark · pandas

Backend & DevOps

FastAPI · PostgreSQL · Redis · Docker · GCP · Terraform · CI/CD (GitHub Actions) · Microservices

Technical Leadership

Software architecture & tech stack decisions · Technical interviewing & hiring · Mentoring · Client-facing technical lead

Languages

Spanish (native) · English C1 (TOEIC 920/990)

EDUCATION

M.Sc. in Data Science

Universitat Oberta de Catalunya (UOC) · 2021 - 2022

B.Sc. in Telecommunications Engineering – Audiovisual & Multimedia Systems

Universidad Rey Juan Carlos (URJC) · 2013 - 2017